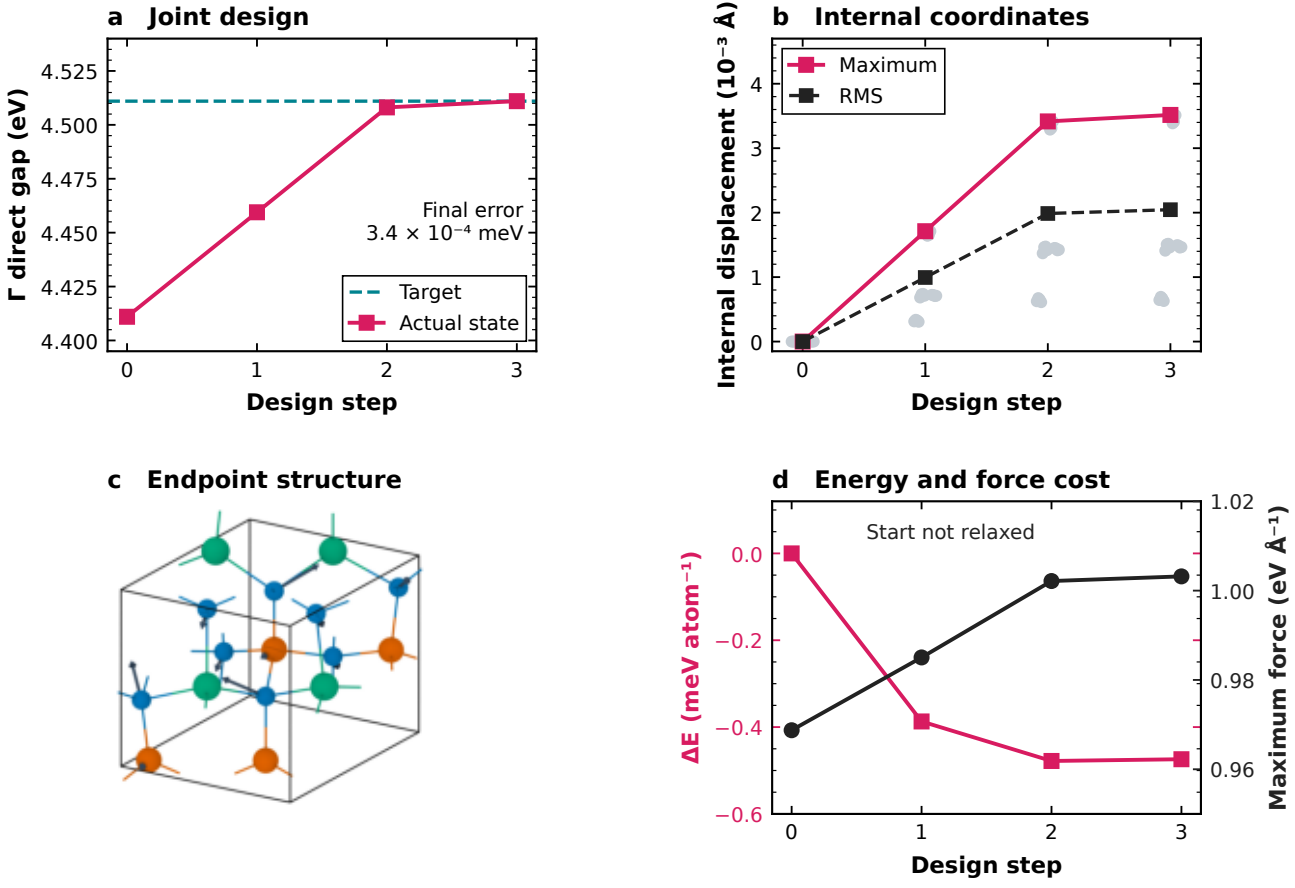
a Joint design
Γ direct gap (eV)
4.525
4.500
4.475
4.450
4.425
4.400
Final error
3.4 × 10⁻⁴ meV
Target
Actual state
0
1
2
3
Design step
b Internal coordinates
Internal displacement (10⁻³ Å)
Maximum
RMS
4
3
2
1
0
0
1
2
3
Design step
c Endpoint structure
d Energy and force cost
Start not relaxed
ΔE (meV atom⁻¹)
0.0
−0.2
−0.4
−0.6
Maximum force (eV Å⁻¹)
1.02
1.00
0.98
0.96
0
1
2
3
Design step